

Response to Reviewer PtZf

Structure-Aware Prediction of PROTAC-Mediated Protein Degradability
via Graph Neural Networks

ACM-BCB 2026 Submission

Rating: 3 — Borderline Reject

We thank Reviewer PtZf for the detailed feedback. We have addressed all concerns and believe the revised manuscript substantially strengthens the empirical evaluation. All changes are [highlighted in blue](#) in the revised paper. Line numbers below refer to the revised `paper.tex`.

W1: Performance is inconsistent across the paper: abstract reports 0.78 AUROC (95% CI 0.74–0.82), main results section reports 0.646 ± 0.124 , appendix reports 5-fold CV mean 0.565.

Response: We believe the “0.78” figure may refer to a prior draft; the current abstract (line 65) reports 0.646 ± 0.124 (multi-seed average). To prevent any confusion, we have added a “Reconciling performance estimates” paragraph that explicitly maps each reported number to its evaluation protocol:

Number	Protocol	Role	Line
0.603 ± 0.097	6-seed, single split	Most conservative	L621
0.646 ± 0.124	3-seed, single split	Primary result	L460
0.7449	Best seed (seed 42)	Peak (secondary)	L462
0.657	Baseline (seed 42)	Single-seed reference	L453
0.565 ± 0.052	5-fold CV	Conservative lower bound	L681

The 6-seed mean (0.603) is recommended for citation as the most conservative estimate. The differences between protocols are now explicitly reconciled:

“We report three evaluation protocols: multi-seed single split (0.646 ± 0.124), baseline single seed (0.657), and 5-fold CV (0.565 ± 0.052 , Appendix Table 9). The 6-seed mean (0.603) is the most conservative and recommended estimate.”

[Paper reference: Lines 680–684 (“Reconciling performance estimates” paragraph in §4.3); Abstract line 65 (primary number clearly stated); Table 1 lines 441–458 (all protocols in one table).]

W2: Robustness is limited. Target-unseen has high seed sensitivity, and 5-fold CV mean is substantially lower than the headline single-split number.

Response: We agree and have substantially expanded the variance analysis:

1. **Extended to 6 seeds** (789, 1011, 1213 added), yielding 0.603 ± 0.097 —marginally below gradient boosting (0.607). [Paper reference: Lines 619–625 (“Six-seed extended validation” paragraph in §4.3).]

2. **Stated explicitly that the improvement is not statistically significant:** “bootstrap $p = 0.259$ ” for 3-seed (line 529) and “ $p = 0.556$ ” for 6-seed (line 621). Previously the p -value appeared only in a figure caption; it is now in the main text. [Paper reference: Lines 461 (“not statistically significant at $\alpha=0.05$ ”), 529 (“bootstrap $p = 0.259$ ”), 621 (“bootstrap $p = 0.556$ ”).]
3. **Recommended ensembling** across ≥ 6 seeds for reliable deployment, noting individual seeds span a 0.25-AUROC range (0.502–0.745). [Paper reference: Line 624 (“model ensembling across ≥ 6 seeds is strongly recommended”).]
4. **Reported both 3-seed and 6-seed** results in Table 1 to show how the estimate changes with more seeds. [Paper reference: Table 1 lines 441–458 (rows for 3-seed and 6-seed).]
5. **Figure 4** (line 632) now visualizes all 6 seeds, distinguishing original (solid) vs. additional (hatched). [Paper reference: Figure 4 at line 632.]
6. **Honest performance assessment** in Discussion: “We emphasize that the 6-seed mean (0.603) is not statistically superior to gradient boosting (0.607, $p = 0.556$).” [Paper reference: Lines 744–748 (“Honest performance assessment” paragraph in §5).]

W3: Validation and model-selection protocol appears weak for the main use case. Validation AUROC does not reliably predict target-unseen test AUROC.

Response: Acknowledged as a methodological limitation. We now discuss this explicitly in the Training Procedure section:

“We observe that validation-split AUROC does not reliably predict target-unseen test AUROC. Ideally, we would use a target-unseen validation split for hyperparameter selection; however, this would further reduce the already-limited training data (155 proteins $\rightarrow$ ~ 120 training, ~ 15 validation, ~ 20 test). We therefore use standard random validation... while acknowledging this may bias hyperparameter selection toward random-split performance.”

We also state the implication: the chosen hyperparameters ($LR=5 \times 10^{-4}$, dropout=0.05, noted at line 323) may not be optimal for target-unseen generalization, contributing to variance.

[Paper reference: Lines 326–330 (“Validation-test mismatch” paragraph in §3.5 Training Procedure); Line 323 (hyperparameter values); Limitation item 4 at line 769 (“Validation-test mismatch: Validation AUROC does not reliably predict target-unseen test AUROC”).]

W4: Generalization across E3 ligases remains limited. VHL performance is poor; rare ligases have too few samples.

Response: Fully addressed. E3 claims are narrowed to CRBN $\leftrightarrow$ VHL transfer throughout. Key additions:

- **Per-E3 Table 5** (lines 651–666): Quantitative breakdown—CRBN 0.758 ($n=273$), VHL 0.396 ($n=187$), cIAP1 0.417 ($n=8$). [Paper reference: Table 5 at line 654.]
- **Per-E3 Figure 5** (lines 668–674): Visual bar chart with random-chance reference line. [Paper reference: Figure 5 at line 673.]

- **VHL root cause analysis** (lines 640–648): Ruled out class imbalance (VHL 36.9% positive vs CRBN 38.4%) and dataset size (91 vs 125 unique proteins). Identified structural heterogeneity—VHL-targeted proteins span diverse protein families with varying surface topology. [Paper reference: Lines 640–648.]
- **Explicit practitioner warning** (line 647): “practitioners should expect substantially lower performance than the overall 0.646 mean when deploying on VHL-targeted proteins, and near-random performance for rare E3s.” [Paper reference: Line 647.]
- **E3-unseen vs target-unseen paradox explained** (lines 636–638): VHL can succeed at E3-level transfer (0.811) while failing at protein-level generalization (0.396)—different generalization axes. [Paper reference: Lines 636–638.]
- **Narrowed throughout**: Abstract line 65 (“CRBN→VHL”), section heading line 635 (“limited to major ligases”), conclusion line 783. [Paper reference: Lines 65, 635, 783.]

W5: No external validation reduces confidence in real-world generalization.
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Response: Stated as Limitation item 6:

“No external validation: PROTAC-8K is the only publicly available benchmark with sufficient scale; generalization to data from other assay platforms or research groups is unvalidated.”

We also caution against deployment without prospective validation on independently collected data (Discussion lines 744–748, Conclusion lines 783–784).

[Paper reference: Limitation item 6 at line 771; Discussion “Honest performance assessment” lines 744–748; Conclusion lines 783–784 (“neither is statistically superior to gradient boosting”).]

Summary of All Changes Relevant to Reviewer PtZf

Concern	Revision (lines in paper.tex)
Inconsistent reporting (W1)	§4.3 L680–684 (reconciliation); Abstract L65; Table 1 L441–458
Limited robustness (W2)	§4.3 L619–625 (6-seed); Table 1 L452; Figure 4 L632; <i>p</i> -values L461, L529, L621; L624 (ensembling); §5 L744–748
Weak validation protocol (W3)	§3.5 L326–330; L323; Limitation 4 L769
Limited E3 generalization (W4)	Table 5 L654; Figure 5 L673; Root cause L640–648; Warning L647; Paradox L636–638; Abstract L65; Heading L635; Conclusion L783
No external validation (W5)	Limitation 6 L771; §5 L744–748; Conclusion L783–784
<i>New analyses added in revision:</i>	
6-seed extended validation	Table 1 L452; §4.3 L619–625; Figure 4 L632
Calibration analysis (ECE = 0.029)	§4.6 L705–730; Table 6 L714
Per-E3 breakdown	Table 5 L654; Figure 5 L673
VHL root cause analysis	§4.3 L640–648
BRD4 case study	§4.5 L686–703
Lysine paradox discussion	§5 L751–758; Figure 7 L762
Statistical non-significance	L461, L529, L621